

There are four problems on this exam worth 25 points each.

Problems

1. Compute the closed form expression for $y[n]$:

$$y[n] = x[n] * h[n],$$

where

$$x[n] = \left(\frac{1}{2}\right)^n u[n], \quad h[n] = u[n].$$

2. Compute $x[n]$ for the $X(\omega)$ shown in Figure 1.

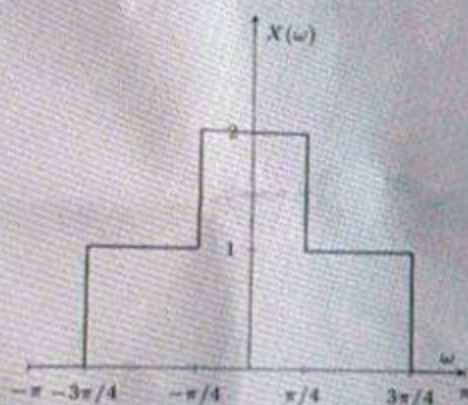


Figure 1: Discrete-time Fourier Transform for Problem 2

3. Compute $x[n]$ for $X(\omega)$ shown in Figure 2.

$$\left[\frac{e^{-\frac{\pi}{4}jn}}{jn} \right] - \left[\frac{e^{-\frac{3\pi}{4}jn}}{jn} \right]$$

$$= \frac{1}{jn}$$

Question # 4

CLO-01[12+5=17 Points]

The values of DTFT of sequence $x(n) = \{a, b, c\}$ for $0 \leq n \leq 2$, at frequencies $\omega = \frac{\pi}{2}$, 3π and 6π are given by 3-j, 0, and 2 respectively.

- Determine the values of samples a, b, c.
- What is the advantage of Discrete Time Fourier Series over Continuous Time Fourier Series?

Question # 5

CLO-02 [10+10=20 Points]

Consider the sequence $x(n) = 5\delta(n) + 2\delta(n-2) + 3\delta(n-3)$. If $X(k)$ is the 4 point DFT of $x(n)$

- Find the finite sequence $y_1(n)$ if $Y_1[k] = W_4^{-3k} X[k]$.
- The four-point DFT of $x(n)$ is computed and the resulting sequence is squared such that $Y_2[k] = X^2[k]$. A four-point inverse DFT is then computed to produce sequence $y_2(n)$. Find the sequence $y_2(n)$.

Question # 6

[5+10+5=20 Points]

- Find all possible $x(n)$ associated with the Z-transform:

$$X(z) = \frac{5z^{-1}}{(3-z^{-1})(1-2z^{-1})}$$

- A causal linear shift-invariant system is characterized by the difference equation

$$y(n) = \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) + x(n) - x(n-1)$$

Find the system function $H(z)$, and the unit sample response $h(n)$, and determine whether or not system is stable.

- The z-transform of a sequence $x(n)$ is:

$$X(z) = \frac{1-4z^{-1}+2z^{-2}}{1-3z^{-1}+0.5z^{-2}}$$

Find the DTFT of $x(n)$ at $\omega = \pi/2$

Formulae:

$$\text{DFT: } X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} \quad k = 0, 1, 2, \dots, N-1 \quad \text{IDFT: } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi kn}{N}} \quad n = 0, 1, 2, \dots, N-1$$

$$\text{IDTFS: } x(n) = \sum_{k=0}^{N-1} c_k e^{j\frac{2\pi kn}{N}}$$

$$\text{DTFS: } c_k = \frac{1}{N} \sum_{n=0}^{N-1} X(k) e^{-j\frac{2\pi kn}{N}}$$

$$\text{DTFT: } X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

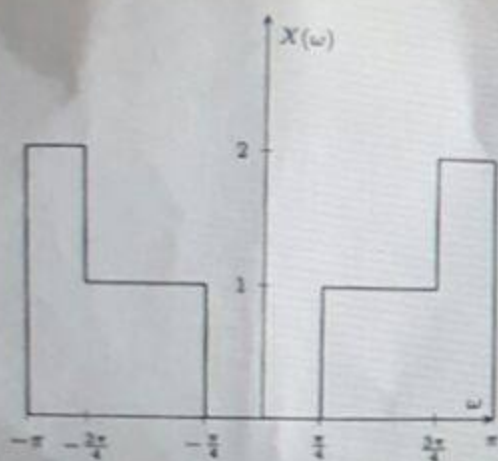


Figure 2: Discrete-time Fourier Transform for Problem 3

4. Compute the Discrete-time Fourier Transform of

$$\frac{\sin(n\pi/2)}{n\pi} \frac{\sin(n\pi/4)}{n\pi}$$

$$e^{-\frac{\pi}{4}jn} - e^{-\frac{3\pi}{4}jn} + 2e^{-\frac{\pi}{4}jn} - 2e^{\frac{\pi}{4}jn} + 2e^{\frac{3\pi}{4}jn} - e^{\frac{\pi}{4}jn}$$

$$\delta[n-n_0] \leftrightarrow e^{-j\omega n_0}$$

$$N_0 = 2\pi t.$$

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Digital Signal Processing	Course Code:	EE302
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	60 Minutes	Total Marks:	45
Paper Date:	20-Feb-17	Weight	15%
Section:	EE	Page(s):	1
Exam:	Mid-I	Roll No:	L14

Instruction/Notes: Please write legibly, as readability and understandability are important during marking. In case of more than one attempts to a question, only one will be evaluated, whichever is found first.

Question # 1 15-Points

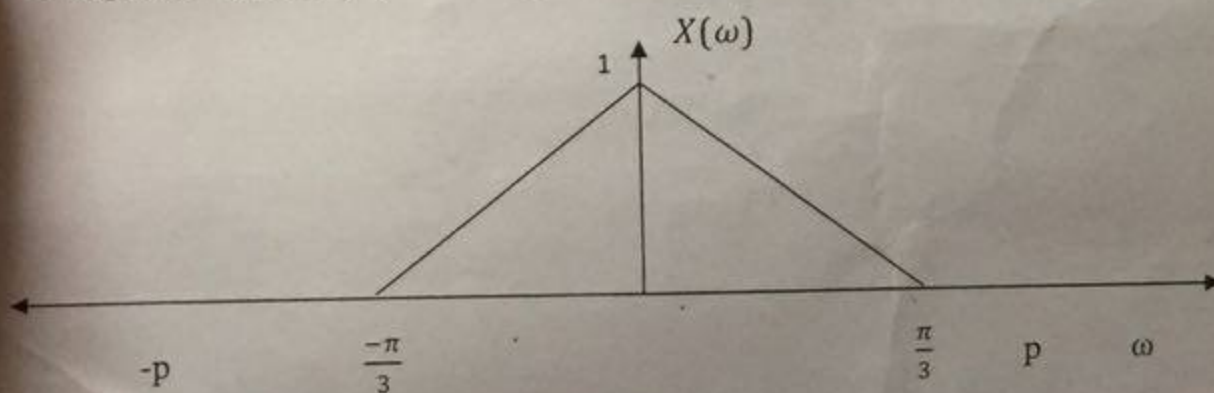
CLO-01

A discrete time periodic signal $x(n) = \{4, 2, 1, 0\}$. Plot its power density spectrum for $-7 \leq k \leq 7$

Question # 2 10-Points

CLO-01

A sequence certain $x(n)$ has zero phase DTFT $X(\omega)$ as sketched in figure below.



Sketch the DTFT $X(\omega)$ of the sequences:

- $x(-n)$
- $x(n) \cos(\frac{\pi}{6}n)$

CLO-02

Question # 3 10-Points

Consider a discrete time a periodic sequence $x(n) = \{1, 2, 3, 4\}$ with its DTFT as $X(\omega)$. We sample $X(\omega)$ by taking 4 equidistant samples in each cycle of $X(\omega)$, i.e., $X(k)$, $k = 0, 1, 2, 3$. Sketch corresponding discrete time signal for $-7 \leq n \leq 7$ against the frequency domain samples.

Hint: You need not to compute $X(k)$.

CLO-02

Question # 4 10-Points

Determine the circular convolution of the sequences:

$$x_1(n) = [1, 2, 3, 1] ; x_2(n) = [4, 3, 2, 2]$$



Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 6-April-17
Section: EE
Exam: Mid-II

Semester: Spring 2017
Total Marks: 50
Weight: 15%
Page(s): 1
Roll No: 44-4435

Instruction/Notes: Please write legibly, as readability and understandability are important during marking. In case of more than one attempts to a question, only one will be evaluated, whichever is found first.

Question # 1

CLO-03

20-Points

If the sequences $x(n) = \{1, 2, 2, 1\}$ and $h(n) = \{3, 4\}$, compute $y(n) = x(n) * h(n)$ using overlap add method while assuming $L = 2$.

Question # 2

CLO-04

10-Points

Draw the direct form II structure for the system described by the following equation:

$$2y(n) + y(n-1) - 4y(n-3) = x(n) + \frac{1}{2}x(n-5)$$

Question # 3

CLO-04

10-Points

For each of the following systems, determine whether the system is (1) causal, (2) time invariant.

(a) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$

(b) $y(n) = x(-n+2)$

Question # 4

CLO-04

10-Points

Determine the Z-transform and sketch the region of convergence (ROC) of the following signal:

$$x(n) = \begin{cases} \left(\frac{1}{3}\right)^n & n \geq 0 \\ \left(\frac{1}{2}\right)^{-n} & n < 0 \end{cases}$$

National University of Computer and Emerging Sciences, Lahore Campus



Course Name:	Digital Signal Processing	Course Code:	EE302
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	180 Minutes	Total Marks:	100
Paper Date:	17 th May 2017	Weight:	50
Section:	ALL	Page(s):	2
Exam Type:	Final		

Student : Name:

Roll No.

Section:

Instruction/Notes:

Please write legibly, as readability and understandability are important during marking. In case of more than one attempts to a question, only one will be evaluated, whichever is found first.

Question # 1

CLO-01 [5+5=10 Points]

An analog signal contains frequencies up to 20 KHz

- What range of sampling frequencies allows exact reconstruction of this signal from its samples?
- If we sample this signal with a sampling frequency $F_s = 8$ KHz, examine what happens to frequency $F = 7$ KHz.

Question # 2

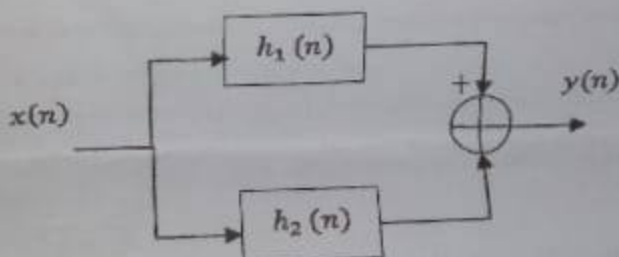
CLO-03 [15 Points]

- Consider the interconnection of LTI system. Find $y(n)$ where

$$x(n) = \{3, 1, 0, -1\}$$

$$h_1(n) = \left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{2} \right\}$$

$$h_2(n) = \{-2, 1, 0\}$$



CLO-06 [5+5+5+3=18 Points]

Question # 3

Consider the filter

$$y(n] = 0.9y[n-1] + bx[n]$$

- Determine b such that the frequency response of the filter satisfies the condition: $|H(0)] = 1$
- Draw pole zero plot of the system function $H(z]$
- Determine the frequency at which $|H(\omega)] = \frac{1}{\sqrt{2}}$
- Would you classify this filter as low pass, high pass, or bandpass? Is this filter stable? Justify your answer.

Problems

1. Consider

$$x[n] = \left(\frac{1}{2}\right)^n u[n],$$

and

$$h[n] = u[n].$$

Compute the closed-form expression of

$$y[n] = x[n] * h[n].$$

2. Consider the finite-length sequence

$$x[n] = \delta[n] + 2\delta[n - 5].$$

(a) Find the 10-point DFT of $x[n]$.

(b) Find the sequence that has a DFT

$$Y(k) = e^{j2k\frac{2\pi}{10}} X(k),$$

where $X(k)$ is the 10-point DFT of $x[n]$.

(c) Find the 10-point sequence $y[n]$ that has a DFT

$$Y(k) = X(k)W(k),$$

where $X(k)$ is the 10-point DFT of $x[n]$, and $W(k)$ is the 10-point DFT of the sequence

$$w[n] = \begin{cases} 1, & 0 \leq n \leq 6 \\ 0 & \text{otherwise.} \end{cases}$$

3. Consider the sequence

$$x[n] = \delta[n] + 2\delta[n - 2] + \delta[n - 3].$$

(a) Find the 4-point DFT of $x[n]$.

(b) If $y[n]$ is the 4-point circular convolution of $x[n]$ with itself, find $y[n]$ and the 4-point DFT $Y(k)$.

(c) With $h[n] = \delta[n] + \delta[n - 1] + 2\delta[n - 3]$, find the 4-point circular convolution of $x[n]$ with $h[n]$.

4. Consider a causal FIR filter of length $M = 6$ with impulse response

$$h[n] = \{2, 2, 2, 2, 2, 2\}.$$

- (a) Provide a closed-form expression for the 8-point DFT of $h[n]$, denoted by $H_8[k]$, as a function of k . Simplify as much as possible.
- (b) Consider the sequence $x[n]$ of length $L = 8$ below, equal to a sum of several finite-length sinewaves:

$$x[n] = \cos\left(\frac{\pi}{2}n\right) + 5\cos(\pi n), \quad n = 0, 1, \dots, 7.$$

$y_8[n]$ is formed by computing $X_8[k]$ as an 8-point DFT of $x[n]$, $H_8[k]$ as an 8-point DFT of $h[n]$, and then $y_8[n]$ as the 8-point inverse DFT of $Y_8[k] = X_8[k]H_8[k]$. Express the result $y_8[n]$ as a weighted sum of finite-length sinewaves similar to how $x[n]$ is written above.

6x7 7x1